MILITARY SPECIFICATION

COMPUTER, BALLISTICS: M13A1

This amendment forms a part of Military Specification MIL-C-45617A, dated 9 November 1962.

Paragraph 3.10.3, line 10: Change "seven other cams" to "eight other cams".

Paragraph 3.16: Delete and substitute:

"3.16 Superelevation cams.- The HVAP, M304, HEP, T142E3 and HEAT, T108 cams shall be placed in the computer and checked, after which these cams shall be removed and placed in the spare cam box. Six cams shall then be placed in the computer in the following sequence starting with the AP-T, M318A1 and following with the HE-T, M71A1; APC-T, M82; HE, M71; HVAP, M332 and HEAT-T, M431. Each computer shall be shipped with the same nine cams with which it qualified."

Table IV: Delete and substitute the attached Tables IV and IVA:

Paragraph 4.3 and 4.3.1: Delete and substitute the following:

"4.3 First article (initial production) approval.- The requirement for first article approval and the responsibility (Government or contractor) for first article testing shall be as specified in the contract. The sample for first article approval tests shall consist of 3 computers plus 3 each of all items covered by SQAP. The sample shall be manufactured in the same manner, using the same materials, equipment, processes, and procedures as used in regular production. All parts and materials, including packaging and packing, shall be obtained from the same source of supply as used in regular production.

4.3.1 Government testing.- When the Government is responsible for conducting first article approval tests, the contractor, prior to submitting the sample to the Government, shall inspect the sample to insure that it conforms to all the requirements of the contract and submit a record of this inspection with the sample, including certificates of conformance for materials.

4.3.2 Contractor testing.- When the contractor is responsible for conducting first article approval tests, the sample shall be inspected by the contractor for all the requirements of the contract. The sample and a record of this inspection, including certificates of conformance for materials, shall be submitted to the Government for approval. The Government reserves the right to witness the contractor s inspection."

Paragraph 4.6.2, line 2: Change "AQAP" to "SQAP".

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Paragraph 4.9.7.2, line 13: Change "table IV" to "table IVA".

Paragraph 4.9.7.2, last sentence: Change "six cams" to "eight cams" and change "table IV" to "tables IV and IVA".

Paragraph 4.9.7.4, line 11: Change "table IV" to "table IVA".

Paragraph 4.9.7.6.1, last sentence following the word specified: Add "in table IVA."

Paragraph 4.9.7.6.2, line 3: Change "table IV" to "tables IV and IVA".

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TABLE IV

Computer accuracy as determined by position of rotation of the output shaft based on 1500 yard reference range for AP-T, M318A1 cam

Range Yards	500	1000	1500	2000	3000	3600	4000	4700	4800
Input shaft Position ± 1°	11R 68.8° cw	4R 46.9° cw	0° Check Point	2R 334.6° ccw	7R 21.6° ccw	8R 330° ccw	9R 356.5° ccw	11R 228° ccw	11R 305° ccw
AP-T, M318A1	1R 89.4° ccw *2.78	232.6° ccw *5.79	0° *9.02	249.5° cw *12.49	2R 87.2° cw *20.24		4R 23.1° cw *29.35		5R 284.2° cw *37.97
HEAT, T108	1R 50.5° ccw *3.32	138.1° ccw *7.11	174.4° cw *11.45	1R 175.4° cw *16.47	4R 7.3° cw *29.13		7R 188.2° cw *46.64		11R 105.1° cw *65.49
HEP, T142E3	1R 3.5° ccw *3.97	23.1° ccw *8.70	1R 33.6° cw *14.50	2R 190.4° cw *21.67	6R 184.2° cw *41.59	9R 294.3° cw *58.11			
HVAP, M332	1R 182.3° ccw *1.49	1R 28.2° ccw *3.63	228.5° ccw *5.85	49.1° ccw *8.34	1R 37.9° cw *14.54		2R 314.3° cw *23.39		4R 333.3° cw *33.66
HE, M71 (Ren)	1R 37.5° ccw *3.50	132.4° ccw *7.18	147.6° cw *11.08	1R 86.3° cw *15.23	3R 25.5° cw *24.38		5R 63.5° cw *34.90		7R 30.6° cw *44.45

*=Corresponding nominal superelevation mil value

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TABLE IV (Continued)

Range Yards	500	1000	1500	2000	3000	3600	4000	4700	4800
APC-T M82	1R 39.3° ccw *3.20	175.4° ccw *6.59	84.6° cw *10.20	1R 2.3° cw *14.06	2R 256.5° cw *22.59		4R 243.2° cw *32.40		6R 165.1° cw *41.32
HVAP M304	1R 124.4° ccw *2.29	302.3° ccw *4.82	101.4° ccw *7.61	119.6° cw *10.69	1R 280.4° cw *17.91	2R 294.9° cw *23.11			
HE-T, M71A1	332.6° ccw *4.4	5.8° cw *9.1	1R 13° cw *14.2	2R 41.8° cw *19.6	4R 200.2° cw *31.8		7R 142.6° cw *46.0		10R 13° cw *59.2
HEAT-T, M431	1R 167° ccw *1.7	1R 8.6° ccw *3.9	174.2° ccw *6.6	85° cw *10.2	2R 193° cw *21.7		7R 164.2° cw *46.3	13R 265° cw *77.7	

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TABLE IV A

A OUTPUT SHAFT ROTATION	B DEGREES	C MILS *
0 up to But less than 6	± 7.20	± 0.10
6 up to But less than 11	± 10.80	± 0.15
11 up to But less than 26	± 14.40	± 0.20
26 up to But less than 31	± 18.00	± 0.25
31 up to But less than 36	± 25.20	± 0.35
36 up to But less than 41	± 28.80	± 0.40
41 up to But less than 61	± 36.00	± 0.50
61 up to But less than 86	± 43.20	± 0.60
86 up to But less than 98.5	± 50.40	± 0.70

*The data contained in Column C is informational only.

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